Koblitz cover transfers — R29

The double image, exact base cancellations and a full-level cubic bridge

2026-10-28 client date / 2026-10-29 UTC publication.

At degree 12 both first adjugates vanish. R29 computes epsilon² coefficients of N(F+epsilon H), epsilon³=0, for H=1, x and 1+x. All original norms match M15 and all first derivatives match R28. Every one of 1,024 coefficients of all three second-order functions is recovered and checked.

With nonzero unweighted coefficient C, the weighted ratios recover the pair's product and sum. At the registered F2^12 support the image polynomial is $X^2 + beta^{11} + beta^8 + beta^5 + beta^3 + beta$. It is a square, selecting exactly one actual curve point, of degree 12 and multiplicity two. This completes the ordinary paired-image data of degrees 460 and 367; it does not identify every original source branch or its colored scalar-C5 lift.

R29: completed image multiplicities and base Frobenius cancellations

Degree 436 × 1	**Frobenius offset: 218**	Base closed divisor cancels
Degree 12 × 2	**Frobenius offset: 6**	One second point, multiplicity 2
Degree 284 × 1	**Frobenius offset: none**	No tested base-deck match
Degree 83 × 1	**Frobenius offset: none**	No tested base-deck match

Ordinary 460 base difference cancels; this does not determine the colored C5 deck offset.

Separate exact check: irreducible degree-3 fiber over F2^14 gives order-77 data over F2^42.

Global independent-model identities, full source lifts and modular-factor columns remain open.

Run IDs: jet29-01, eval29-01, dependencies29-01, cubic29-01. No IC independence claim.

The complete Frobenius/deck checks use actual image fields, ordinary cusp/SS avoidance, a corrupted-ordinate detector, exact rational 2-torsion translation identities and all five coefficient-descent curve/product conjugates.

What cancels, what survives, and the independent model

Degree	Mult.	Frob.	Neg.	T	Neg.+T
436	1	218	−1	−1	−1
12	2	6	−1	−1	−1
284	1	−1	−1	−1	−1
83	1	−1	−1	−1	−1

Here −1 means no match in the complete Frobenius orbit. The ordinary degree-460 base difference cancels exactly as closed divisors on E49. The asymmetric 367 supports do not coincide under these tested base actions. Colored C5 deck offsets and modular-factor columns are still required. Neither a zero nor a positive IC relation rank follows from this table.

CUBIC29 separately uses the actual degree-14 lambda-zero point. The QR line cuts an irreducible, separable cubic. Exact inverse/lift certificates and Tate group laws produce an actual order-77 point over F2^42, with CRT projectors 56/22 and diamond action 23. Monic/DVR specialization then proves a generic degree-three extension. MODULI_BRIDGE.md states the written proper identification with X1(77) and its quotient; native genus arithmetic alone is not that certificate.

The independent Sutherland/van Hoeij model is retrieved and archived with its hash. Its global rational-function match to the incidence coordinates has not been checked. A shared level, genus or finite point is explicitly not substituted for that identity.

All four native runs and the bounded compiler pass. Lean compiles and audits two universal pair-sum/polynomial theorems, zero failures and zero trusted native. Across R27–R29: 40 named theorems, 41 current audited declarations including one generated congruence theorem. The retained-record checker verifies source hashes and 21 accepted native runs among 32 attempts. This is record integrity, not an independent mathematical replay.

The next construction is the colored lift and complete signed divisor, followed by actual modular-factor evaluation. Full old coupled saturation, explicit determinant-19 theta sections and the capped map solve remain open.